

Producer Consumer using Semaphores

```
#include<stdio.h>
#include<pthread.h>
#include<unistd.h>
#include<semaphore.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0, out = 0;
sem_t empty; // counts empty slots
sem_t full; // counts filled slots
pthread_mutex_t mutex; // for mutual exclusion
void* producer(void* arg) {
    int item, i = 0;
    while (1) {
        item = i++; // produce an item (here just incrementing number)
        sem_wait(&empty); // wait for empty slot
        pthread_mutex_lock(&mutex); // enter critical section
        buffer[in] = item;
        printf("Produced: %d at buffer[%d]\n", item, in);
        in = (in + 1) % BUFFER_SIZE;
        pthread_mutex_unlock(&mutex); // leave critical section
        sem_post(&full); // signal that new item is available
        sleep(1); // simulate time to produce
    }
}

void* consumer(void* arg)

{
    int item;
    while (1)
```

```

{
sem_wait(&full); // wait for filled slot
pthread_mutex_lock(&mutex); // enter critical section
item = buffer[out];
printf("Consumed: %d from buffer[%d]\n", item, out);
out = (out + 1) % BUFFER_SIZE;
pthread_mutex_unlock(&mutex); // leave critical section
sem_post(&empty); // signal that a slot is free
sleep(2); // simulate time to consume
}
}
int main()
{
printf("Nitya R Swadi, 1BM24CS361\n");

pthread_t prod, cons;

sem_init(&empty, 0, BUFFER_SIZE);
sem_init(&full, 0, 0); pthread_mutex_init(&mutex, NULL);

pthread_create(&prod, NULL, producer, NULL);
pthread_create(&cons, NULL, consumer, NULL);
pthread_join(prod, NULL);
pthread_join(cons, NULL);

sem_destroy(&empty);
sem_destroy(&full);

pthread_mutex_destroy(&mutex);
return 0;

}

```

OUTPUT:

C:\Users\Nitya\Downloads\rr X

+ v

```
Nitya R Swadi, 1BM24CS361
Produced: 0 at buffer[0]
Consumed: 0 from buffer[0]
Produced: 1 at buffer[1]
Consumed: 1 from buffer[1]
Produced: 2 at buffer[2]
Produced: 3 at buffer[3]
Consumed: 2 from buffer[2]
Produced: 4 at buffer[4]
Produced: 5 at buffer[0]
Consumed: 3 from buffer[3]
Produced: 6 at buffer[1]
Produced: 7 at buffer[2]
Consumed: 4 from buffer[4]
Produced: 8 at buffer[3]
Produced: 9 at buffer[4]
Consumed: 5 from buffer[0]
Produced: 10 at buffer[0]
Consumed: 6 from buffer[1]
Produced: 11 at buffer[1]
Consumed: 7 from buffer[2]
Produced: 12 at buffer[2]
|
```